Diagnosing Why a Student Is Wrong Under Item Budgets

A Confirmatory Evaluation Protocol for Prerequisite-Probe Timing in High-School Chemistry

Manuscript status: confirmatory result binding complete; H1-H4 were evaluated and H5 was excluded pre-outcome with machine-bound evidence. All result values and decisions in the generated sections come from the validated machine contract.

Abstract

An adaptive diagnostic system can often estimate whether a student will answer a target-concept question correctly. A harder problem is to distinguish *why* the student is likely to fail. YHer represents a student's state at one chemistry concept as mastered (M), prerequisite gap (P), chain instability (C), or unlearned target knowledge (U). Under the current binary response channel, local target questions make P and U only weakly distinguishable within short sessions: their production correct-response probabilities differ by 0.10. They are nevertheless asymptotically identifiable under the model, so this work makes no claim of structural, mathematical, or asymptotic non-identifiability. It studies **budget-limited weak identifiability**, or **practical non-identifiability at the pre-specified budgets of 9, 15, and 25 items**.

A hypothesis-generating T0 pilot on synthetic difficulty grids suggested that prerequisite questions could rescue P-state diagnosis, while inserting them at a fixed quota could increase C-state misdiagnosis. The confirmatory design therefore compares three policies on the trusted production catalog and unmodified production inference engine: belief-triggered expected information gain (A), a fixed local-item difficulty ladder (B), and the same ladder with a prerequisite item at every third position (C). It pairs response noise and held-out outcomes across arms, reports matched and deliberately misspecified response generators separately, and freezes analysis populations, stopping rules, seeds, outcomes, and hypothesis decisions before collection. A secondary, separately gated study uses LLM-simulated personas; it is not evidence about human learners. Confirmatory conclusions are admitted only through the machine-generated results contract.

Confirmatory findings. H1=partially_supported (complete); H2=not_supported (complete); H3=supported (complete); H4=supported (complete); H5=not evaluated (excluded pre-outcome). H1 A rate=0.128696; rescue 95% CI=[0.103478, 0.138261]. H2 harm 95% CI=[0.023478, 0.089565]; no-harm 95% CI=[0.072174, 0.122609]. These are simulated finite-budget results, not evidence of human learning or educational efficacy.

1. Introduction

1.1 From correctness to cause

Educational software commonly observes answers, scores them, and recommends the next item. That loop can still fail a student if the same wrong answer is compatible with different causes. A student may lack a prerequisite, understand the components but lose the reasoning chain, or have little target knowledge at all. Those causes imply different next actions. A prerequisite gap calls for an earlier concept; chain instability calls for a worked causal bridge; an unlearned concept calls for a direct introduction.

YHer is a pre-alpha, local high-school chemistry system built around this distinction. Its narrow product loop is: diagnose one concept, route to an existing teacher video or constrained explanation, administer independent practice and held-out questions, and update an event-reconstructable student profile. The system does not claim learning gains, long-term efficacy, or human validation. This paper isolates one measurement question inside that loop: under a fixed item budget and a binary response channel, when should the system ask a prerequisite question?

1.2 The measurement problem

For a target-local item of difficulty d, the production model uses the state-wise correct-response vector

$$P(Y = 1 \mid M, P, C, U) = (0.90,\ \gamma + 0.10,\ 0.70 - 0.20d,\ \gamma),$$

where $\gamma = 0.25$ for multiple-choice items and 0.03 for numeric items. P and U therefore differ locally by 0.10. Given unlimited independent observations under the model, that difference is enough for asymptotic identification. Under 9, 15, or 25 observations, however, posterior evidence may be too weak to produce a correct, confident P classification. A prerequisite item uses a different likelihood vector,

$$P(Y_{pre} = 1 \mid M, P, C, U) = (0.90,\ \gamma,\ 0.80,\ 0.75),$$

and can separate P from U more strongly. The same item may be harmful when asked of a true C student because C, M, and U all have high prerequisite-correct probabilities. The research question is therefore about *probe timing*, not whether prerequisite items are universally useful.

1.3 Research questions and scope

The primary questions are:

1. At the pre-specified budgets, does a belief-triggered prerequisite policy improve correct P convergence over a local-only fixed ladder?
2. Does fixed-quota prerequisite insertion increase C-state misdiagnosis relative to belief-triggered selection, while belief-triggered selection avoids material harm relative to the local-only ladder?

Adaptive item selection itself is established prior art. The intended contribution is not a new CAT algorithm. It is a pre-data, production-bound evaluation of whether a four-cause diagnostic state can be distinguished under limited binary evidence, and whether two prerequisite-probe timing policies have different failure profiles.

1.4 Contributions

This work contributes:

- a bounded four-state diagnostic formulation tied to an operational chemistry loop;
- a three-arm comparison that separates the value of a probe from the policy that schedules it;
- paired held-out and misspecification protocols that expose internal calibration and model-dependence rather than hiding them;
- a frozen, hash-bound execution and analysis contract in which negative results are reportable outcomes; and
- a secondary LLM-persona protocol with manipulation checks and explicit separation from claims about human students.

2. Related Work

2.1 Computerized adaptive testing

Tailored and computerized adaptive testing long predates this project. Lord's Robbins-Monro formulation, Weiss's work on measurement efficiency, and Weiss and Kingsbury's educational applications establish the core premise that item selection can improve measurement efficiency (Frederic M. Lord 1971; David J. Weiss 1982; David J. Weiss and G. Gage Kingsbury 1984). Simulation is also a standard method for examining CAT item selection rules (Juan Ramón Barrada et al. 2010; Kyung (Chris) Tyek Han 2018). YHer therefore treats adaptive selection and simulation as prior art. Its narrower question concerns a causal state distinction that depends on cross-node prerequisite evidence and may remain weak within a fixed budget even when each arm uses legitimate item selection.

2.2 Learner modeling and knowledge tracing

Bayesian knowledge tracing models the acquisition of procedural knowledge over time (Albert T. Corbett and John R. Anderson 1994), and later work compares BKT with logistic and related learner models (Radek Pelánek 2017). YHer borrows Bayesian updating, but the confirmatory study is not a knowledge-acquisition model. It holds one latent truth state fixed during a single diagnostic journey and asks whether an observation policy can recover it. The study therefore does not validate longitudinal learning tracking, transition parameters, or a general student model.

2.3 Spaced repetition

Spaced-repetition research studies review scheduling and memory dynamics (Behzad Tabibian et al. 2019; Junyao Ye et al. 2022). YHer's product profile contains an **FSRS-inspired** stability and decay projection, informed operationally by the official FSRS software repository and Anki documentation (Open Spaced Repetition contributors, n.d.; Anki contributors, n.d.). FSRS is software rather than a peer-reviewed source in this bibliography, and neither the software nor the cited scheduling papers validate YHer's particular decay formula. The confirmatory experiment starts from a uniform belief and advances simulated time only within one session; it does not evaluate spaced-repetition efficacy.

2.4 LLM-simulated students

Recent work uses LLM-simulated students to evaluate questions, tutoring agents, and behavior across learner profiles (Xinyi Lu and Xu Wang 2024; Zhengyuan Liu et al. 2024; Hyoungwook Jin et al. 2025; Tao Wu et al. 2025; Alexander Scarlatos et al. 2025). Work directly questioning whether simulated tutoring behavior has substance rather than surface plausibility reinforces the need for validation gates (Alexander Scarlatos et al. 2026). In this study, LLM personas are secondary. They must pass declared accuracy-band and misconception-level manipulation checks, and provider cells must meet completion thresholds. Even a successful LLM-persona analysis would show cross-provider behavior under prompts, not human validity or educational efficacy.

All citation metadata and verified links are stored in `references.json`.

3. System

3.1 Product boundary

YHer's current wedge is Shanghai high-school chemistry. The knowledge graph contains 135 nodes, but the trusted deterministic catalog opens 27 nodes for the diagnostic loop. The S0 prerequisite census mechanically admits 23 of those nodes to the H1/H2 analysis set and records four prerequisite-free nodes as exclusions from those two hypotheses. The full descriptive and H3 grid retains all 27.

The source pool is gated by the R5 service ledger. A question must also have a trusted mapping, a verified answer, deterministic scoring, no required unresolved media, and a content-family identity before it enters the confirmatory catalog. Two local families per target are reserved for held-out scoring and removed from every administered pool.

3.2 Five production engines

The system separates five concerns:

1. **Mastery** maintains the M/P/C/U posterior, applies production likelihoods, and contains the FSRS-inspired memory projection used outside the static experiment.
2. **Selector** scores eligible items by expected information gain (EIG), applies holdout and seen-item gates, and calls the production stopping rule.
3. **Planner** maps 30-, 60-, and 120-minute product modes to bounded diagnostic, learning, and verification work. The 30-minute product mode is explicitly shallow.
4. **Recommender** ranks existing video segments using profile, purpose, diagnosis, content fit, signed track metadata, and seen-segment exclusions.
5. **Memory** admits selected high-value events and supports reconstruction from the append-only event history.

The confirmatory runner imports the production mastery and selector modules. It does not reimplement Bayesian updating, EIG, or convergence.

3.3 Evidence separation

The product service freezes diagnostic, practice, and held-out partitions at both item and content-family levels. Scoring keys remain server-side. Held-out responses are independent evidence in the product; in the programmatic study, held-out outcomes are paired across arms and never passed into posterior updating. Matched held-out Brier is an internal calibration check only.

4. Hypothesis-Generating Pilot

T0 was run on 2026-07-08 before the production-bound confirmatory study. It used synthetic difficulty grids, a uniform prior, 4,000 simulated students per cell, and the then-specified production likelihood constants. It did not use the real R5 item distribution or the final production selector. Its purpose was to reveal failure modes and generate hypotheses; its values are not pooled with confirmatory estimates.

In the binary local-item condition, C-state correct convergence at nine items ranged from 13.9% to 34.7% across the pilot difficulty settings, reached 45.1% to 61.3% at 15 items, and reached 58.4% to 70.1% at 25. For P at medium difficulty, correct convergence was 0.0% at both 9 and 15 items in that pilot. Replacing three of nine local items with fixed-position prerequisite items increased pilot P correct convergence from 0.0% to 73.9%. The same fixed quota increased C misdiagnosis from 26.8% to 46.1% while leaving C correct convergence nearly unchanged.

These pilot observations motivated H1 and H2. They do not prove either hypothesis. The pilot used a fixed prerequisite quota as a stress comparison, not the final belief-triggered policy, and its idealized generator shared important assumptions with inference. The source report and raw pilot table are `/Users/mac/Desktop/项目文件夹/Tools/scratchpad/T0_dosimetry_report.md` and `/Users/mac/Desktop/项目文件夹/Tools/scratchpad/t0_results.json`.

5. Confirmatory Method

5.1 Frozen hypotheses

- **H1, P-state rescue:** at budget 15 under the matched generator, Arm A reaches at least 0.50 P correct convergence and the 95% confidence interval for A minus B lies strictly above zero. The ordered partial/not-supported branches are frozen.
- **H2, probe-policy harm:** at budget 9, the 95% interval for C minus A C-state misdiagnosis lies above zero, while the upper interval bound for A minus B lies below the +0.05 no-harm margin.
- **H3, adaptive sanity check:** Arm A is compared with B on overall terminal accuracy and convergence time. This is subordinate because adaptive testing is prior art.
- **H4, misspecification:** the H1 rescue and H2 harm point directions are checked under the frozen misspecified generator, with all degradation reported.
- **H5, LLM personas:** provider coverage, weak/strong accuracy bands, and a misconception-hit contrast determine whether secondary cross-provider behavior is reportable.

The exact ordered decision rules are canonical in `experiments/analysis_plan.md`.

5.2 Grid and pairing

The programmatic intention-to-simulate grid contains 27 targets, four truth states, three arms, 50 replicates, and two generator conditions: 32,400 journeys. One maximum-25-item trajectory supplies views at nominal budgets 9, 15, and 25. When a journey reaches confidence early, its posterior and convergence time carry forward without inflating the actual administered count.

A paired unit is target x truth x generator condition x replicate. Arms share the response-noise stream. The two held-out family outcomes are generated without an arm component in their seeds, so all three arms receive the same held-out outcomes.

5.3 Arms

- **A, belief-triggered EIG:** calls production `selector.select_next` at every administered decision. Eligible prerequisite candidates compete with local items.
- **B, fixed local ladder:** cycles requested difficulties 0.25, 0.50, 0.75, and 1.00 and serves local items only.
- **C, fixed quota:** uses B's ladder but replaces positions 3, 6, 9, …, 24 with prerequisite items.

Fixed-arm ties use `(absolute difficulty distance, family_id, item_id)`. All arms use family-epoch replenishment after exhausting unseen eligible families; for B and C, the frozen fixed-selection tuple continues to govern choices within that constraint. Every arm calls production `mastery.observe` for each observation and production `selector.should_stop` after each update. At item 25, a second call with budget 26 separates confidence convergence from budget exhaustion.

5.4 Generators and inference separation

The matched generator samples from production response probabilities. The misspecified generator draws item-level slip from U[0.05, 0.20], guess from U[0.15, 0.35] for every item type, and one replicate-level Normal(0, 0.05) ability offset clipped to [-0.10, 0.10]. Inference keeps the unmodified production constants. Generator parameters never enter the likelihood update.

The misspecified arm reduces, but cannot eliminate, circularity between the simulator and inference model. It remains a synthetic stress test and is reported separately from matched results and by item type.

5.5 Analysis sets and sparse pools

H1/H2 primary estimates use the mechanically eligible 23-target set. The required empirical with-replacement stress estimand preserves all eligible targets and reports exact-item and family repeats. A second, pre-outcome sensitivity estimand uses a budget-specific common-support set on which all three arms can complete without family repetition. Neither result may be suppressed because it is less favorable.

5.6 Outcomes and intervals

At each nominal budget, the analysis reports correct convergence, converged misdiagnosis, terminal argmax accuracy, a four-state confusion matrix, convergence time, held-out Brier, actual item count, and repeat metrics. Severe M-to-U or U-to-M misdiagnosis is reported over all terminal outputs and separately over converged outputs.

Intervals use exactly 10,000 target-stratified percentile bootstrap resamples. Within each target, replicate IDs are resampled with replacement while paired arms remain paired. Contrasts are computed inside each resample. The paper reports numerators, denominators, effects, and intervals rather than relying on a naked p-value.

5.7 Isolation and provenance

Every raw event, journey, shard manifest, and run manifest is marked simulated and contains a persona ID, provider, and model ID. The runner binds an annotated tag, runner commit, current and historical analysis-plan commits, canonical config hash, seeds, engine and input hashes, and a truthful run-start timestamp. Data execution requires bytecode writes to be disabled from process start. A shard becomes resumable only after a protected-filesystem guard attests that designated paths were unchanged during that computation window. These per-shard attestations do not establish global byte identity across the full wall-clock run: a concurrent test wrote `.pytest_cache` between batches. The isolation evidence therefore supports no protected-path writes during each guarded computation window, not continuous zero-write across the entire interval.

6. Results Contract

The table and audit records below are generated directly from the validated results contract. No confirmatory value in this section is hand-entered.

Machine exclusion disclosure. 1,600 intended journeys were excluded from every estimand. Reasons: `structural_failure` (1,600). Arms: Arm C (1,600). Affected targets: 化学计量（摩尔/阿伏伽德罗）, 原子结构, 基本操作, 物质分类.

Machine post-collection static audit policy. The independently reviewed conditional-metric rule records undefined target/draw denominators as NA, uses no denominator redraw, and discloses attempted and defined bootstrap iterations. This clarification was adopted from static review, not from any result direction; the binder verified policy artifact `static_audit_policy.json`.

H5 excluded-cell disclosure. 12 excluded provider cells and 464 excluded persona cells.

H5 provider lifecycle disclosure. Of the six frozen providers: invalid calibration schema=3; invalid provider artifact=0; missing=0; missing required revision=0; network interruption=1; model-drift exclusion=0; provider-configuration exclusion=0; pre-outcome design exclusion=0; technical interruption=0; post-calibration exclusion=2; other collected=0. Invalid, missing, interrupted, and post-calibration-excluded providers do not enter qualifying-provider metrics. Immutable evidence is bound in `h5/h5_results.json`.

H5 provider identities by lifecycle. invalid calibration schema: `deepseek`, `glm`, `kimi`; network interruption: `doubao`; post calibration exclusion: `minimax`, `tongyi`. Immutable evidence is bound in `h5/h5_results.json`.

Machine item-type generator diagnostic. Because administered journeys are mixed trajectories, these probability-gap summaries are generator diagnostics, not item-type H1/H2 outcome estimands. MCQ gap=-0.005756; 95% CI [-0.007220, -0.004277]; 152,853 events / 15,400 journeys / 27 targets. Numeric gap=0.082579; 95% CI [0.080117, 0.085107]; 5,238 events / 3,694 journeys / 13 targets.

Hypothesis	Analysis status	Decision	Machine branch
H1	complete	partially_supported	`mixed`
H2	complete	not_supported	`a_inferior`
H3	complete	supported	`supported`
H4	complete	supported	`supported`
H5	excluded pre-outcome	not evaluated	`excluded_pre_outcome`

Machine display records

H1. - result `H1_P_A_CORRECT_CONVERGENCE_MATCHED_B15_ELIGIBLE_STRESS` / registry `p_rescue.full.matched.b15.arm_A`: value=0.128696; ci95=[0.112174, 0.145217]; numerator=148; denominator=1150; n_target=23; n_pair=1150. - result `H1_P_B_CORRECT_CONVERGENCE_MATCHED_B15_ELIGIBLE_STRESS` / registry `p_rescue.full.matched.b15.arm_B`: value=0.007826; ci95=[0.003478, 0.013043]; numerator=9; denominator=1150; n_target=23; n_pair=1150. - result `H1_P_A_MINUS_B_CORRECT_CONVERGENCE_MATCHED_B15_ELIGIBLE_STRESS` / registry `h1.primary.matched.b15.rescue_A_minus_B`: value=0.120870; ci95=[0.103478, 0.138261]; numerator=139; denominator=1150; n_target=23; n_pair=1150. - result `H1_P_A_MINUS_B_CORRECT_CONVERGENCE_MATCHED_B15_ELIGIBLE_STRESS_CI95` / registry `h1.primary.matched.b15.rescue_A_minus_B`: value=[0.103478, 0.138261]; numerator=139; denominator=1150; n_target=23; n_pair=1150. - result `H1_P_A_CORRECT_CONVERGENCE_MATCHED_B15_COMMON_SUPPORT` / registry `h1.no_repeat.matched.b15.arm_A`: value=0.005000; ci95=[0, 0.015000]; numerator=1; denominator=200; n_target=4; n_pair=200. - result `H1_P_B_CORRECT_CONVERGENCE_MATCHED_B15_COMMON_SUPPORT` / registry `h1.no_repeat.matched.b15.arm_B`: value=0; ci95=[0, 0]; numerator=0; denominator=200; n_target=4; n_pair=200. - result `H1_P_A_MINUS_B_CORRECT_CONVERGENCE_MATCHED_B15_COMMON_SUPPORT` / registry `h1.no_repeat.matched.b15.rescue_A_minus_B`: value=0.005000; ci95=[0, 0.015000]; numerator=1; denominator=200; n_target=4; n_pair=200. - result `H1_P_A_MINUS_B_CORRECT_CONVERGENCE_MATCHED_B15_COMMON_SUPPORT_CI95` / registry `h1.no_repeat.matched.b15.rescue_A_minus_B`: value=[0, 0.015000]; numerator=1; denominator=200; n_target=4; n_pair=200.

H2. - result `H2_C_A_MISDIAGNOSIS_MATCHED_B9_ELIGIBLE_STRESS` / registry `c_misdiagnosis.full.matched.b9.arm_A`: value=0.373913; ci95=[0.346087, 0.401739]; numerator=430; denominator=1150; n_target=23; n_pair=1150. - result `H2_C_B_MISDIAGNOSIS_MATCHED_B9_ELIGIBLE_STRESS` / registry `c_misdiagnosis.full.matched.b9.arm_B`: value=0.276522; ci95=[0.250435, 0.302609]; numerator=318; denominator=1150; n_target=23; n_pair=1150. - result `H2_C_C_MISDIAGNOSIS_MATCHED_B9_ELIGIBLE_STRESS` / registry `c_misdiagnosis.full.matched.b9.arm_C`: value=0.430435; ci95=[0.401739, 0.459130]; numerator=495; denominator=1150; n_target=23; n_pair=1150. - result `H2_C_A_MISDIAGNOSIS_MATCHED_B9_COMMON_SUPPORT`

/ registry `h2.no_repeat.matched.b9.arm_A`: value=0.417778; ci95=[0.373333, 0.462222]; numerator=188; denominator=450; n_target=9; n_pair=450. - result `H2_C_B_MISDIAGNOSIS_MATCHED_B9_COMMON_SUPPORT` / registry `h2.no_repeat.matched.b9.arm_B`: value=0.284444; ci95=[0.242222, 0.326667]; numerator=128; denominator=450; n_target=9; n_pair=450. - result `H2_C_C_MISDIAGNOSIS_MATCHED_B9_COMMON_SUPPORT` / registry `h2.no_repeat.matched.b9.arm_C`: value=0.437778; ci95=[0.393333, 0.482222]; numerator=197; denominator=450; n_target=9; n_pair=450. - result `H2_C_C_MINUS_A_MISDIAGNOSIS_MATCHED_B9_ELIGIBLE_STRESS` / registry `h2.primary.matched.b9.harm_C_minus_A`: value=0.056522; ci95=[0.023478, 0.089565]; numerator=65; denominator=1150; n_target=23; n_pair=1150. - result `H2_C_C_MINUS_A_MISDIAGNOSIS_MATCHED_B9_ELIGIBLE_STRESS_CI95` / registry `h2.primary.matched.b9.harm_C_minus_A`: value=[0.023478, 0.089565]; numerator=65; denominator=1150; n_target=23; n_pair=1150. - result `H2_C_A_MINUS_B_MISDIAGNOSIS_MATCHED_B9_ELIGIBLE_STRESS` / registry `h2.primary.matched.b9.no_harm_A_minus_B`: value=0.097391; ci95=[0.072174, 0.122609]; numerator=112; denominator=1150; n_target=23; n_pair=1150. - result `H2_C_A_MINUS_B_MISDIAGNOSIS_MATCHED_B9_ELIGIBLE_STRESS_CI95` / registry `h2.primary.matched.b9.no_harm_A_minus_B`: value=[0.072174, 0.122609]; numerator=112; denominator=1150; n_target=23; n_pair=1150. - result `H2_C_C_MINUS_A_MISDIAGNOSIS_MATCHED_B9_COMMON_SUPPORT` / registry `h2.no_repeat.matched.b9.harm_C_minus_A`: value=0.020000; ci95=[-0.033333, 0.075556]; numerator=9; denominator=450; n_target=9; n_pair=450. - result `H2_C_C_MINUS_A_MISDIAGNOSIS_MATCHED_B9_COMMON_SUPPORT_CI95` / registry `h2.no_repeat.matched.b9.harm_C_minus_A`: value=[-0.033333, 0.075556]; numerator=9; denominator=450; n_target=9; n_pair=450. - result `H2_C_A_MINUS_B_MISDIAGNOSIS_MATCHED_B9_COMMON_SUPPORT` / registry `h2.no_repeat.matched.b9.no_harm_A_minus_B`: value=0.133333; ci95=[0.091111, 0.175556]; numerator=60; denominator=450; n_target=9; n_pair=450. - result `H2_C_A_MINUS_B_MISDIAGNOSIS_MATCHED_B9_COMMON_SUPPORT_CI95` / registry `h2.no_repeat.matched.b9.no_harm_A_minus_B`: value=[0.091111, 0.175556]; numerator=60; denominator=450; n_target=9; n_pair=450.

H3. - result `H3_A_MINUS_B_TERMINAL_ACCURACY_MATCHED_B15_FULL_SET` / registry `h3.matched.b15.terminal_accuracy_A_minus_B`: value=0.100370; ci95=[0.088889, 0.111852]; numerator=542; denominator=5400; n_target=27; n_pair=5400. - result `H3_A_MINUS_B_TERMINAL_ACCURACY_MATCHED_B15_FULL_SET_CI95` / registry `h3.matched.b15.terminal_accuracy_A_minus_B`: value=[0.088889, 0.111852]; numerator=542; denominator=5400; n_target=27; n_pair=5400. - result `H3_A_MEDIAN_CONVERGENCE_ITEMS_MATCHED_B15_FULL_SET` / registry `h3.matched.b15.time_to_confidence.arm_A`: value=8; ci95=[8, 8]; numerator=50613; denominator=5400; n_target=27; n_pair=5400. - result `H3_B_MEDIAN_CONVERGENCE_ITEMS_MATCHED_B15_FULL_SET` / registry `h3.matched.b15.time_to_confidence.arm_B`: value=11; ci95=[10, 11]; numerator=55139; denominator=5400; n_target=27; n_pair=5400.

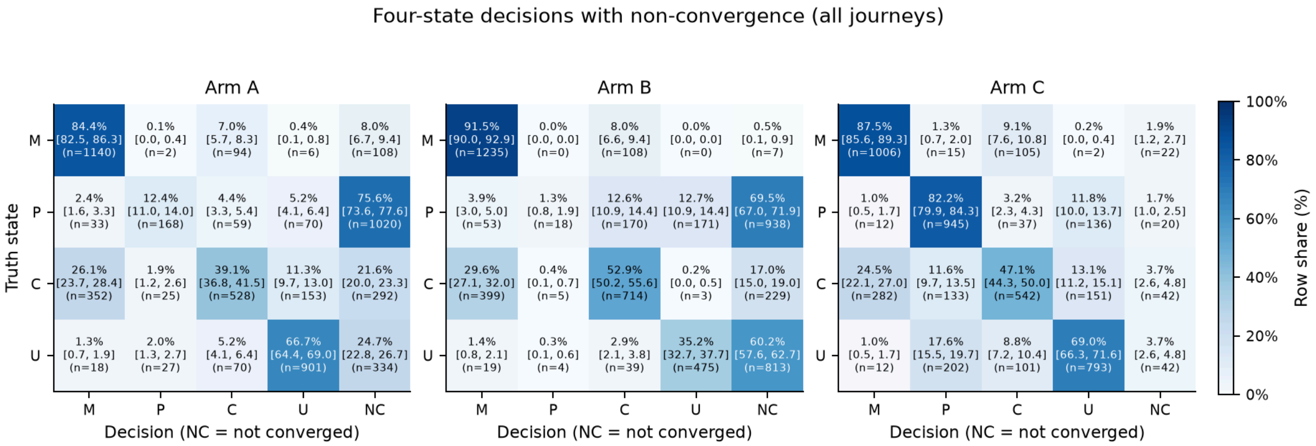
H4. - result `H4_H1_RESCUE_MISSPECIFIED_B15_ELIGIBLE_STRESS` / registry `h4.misspecified.b15.rescue_A_minus_B`: value=0.120870; ci95=[0.101739, 0.140000]; numerator=139; denominator=1150; n_target=23; n_pair=1150. - result `H4_H2_HARM_MISSPECIFIED_B9_ELIGIBLE_STRESS` / registry `h4.misspecified.b9.harm_C_minus_A`: value=0.069565; ci95=[0.036522, 0.102609]; numerator=80; denominator=1150; n_target=23; n_pair=1150. - result `H4_H1_MATCHED_TO_MISSPECIFIED_DEGRADATION` / registry `h4.degradation.h1_rescue.matched_minus_misspecified`: value=0.000000; ci95=[-0.026087, 0.025217]; numerator=0; denominator=1150; n_target=23; n_pair=1150. - result `H4_H2_MATCHED_TO_MISSPECIFIED_DEGRADATION` / registry `h4.degradation.h2_harm.matched_minus_misspecified`: value=-0.013043; ci95=[-0.060000, 0.034783]; numerator=-15; denominator=1150; n_target=23; n_pair=1150. - result `H4_H2_NO_HARM_MISSPECIFIED_B9_ELIGIBLE_STRESS` / registry `h4.misspecified.b9.no_harm_A_minus_B`: value=0.080870; ci95=[0.054783, 0.106957]; numerator=93; denominator=1150; n_target=23; n_pair=1150. - result `H4_H2_NO_HARM_MATCHED_TO_MISSPECIFIED_DEGRADATION` / registry `h4.degradation.h2_no_harm.matched_minus_misspecified`: value=0.016522; ci95=[-0.020870, 0.053043]; numerator=19; denominator=1150; n_target=23; n_pair=1150. - result `H4_H1_RESCUE_MISSPECIFIED_B15_COMMON_SUPPORT` / registry `h1.no_repeat.misspecified.b15.rescue_A_minus_B`: value=0.005000; ci95=[0, 0.015000]; numerator=1;

denominator=200; n_target=4; n_pair=200. - result `H4_H1_RESCUE_MISSPECIFIED_B15_COMMON_SUPPORT_CI95` / registry `h1.no_repeat.misspecified.b15.rescue_A_minus_B`: value=[0, 0.015000]; numerator=1; denominator=200; n_target=4; n_pair=200. - result `H4_H2_HARM_MISSPECIFIED_B9_COMMON_SUPPORT` / registry `h2.no_repeat.misspecified.b9.harm_C_minus_A`: value=0.028889; ci95=[-0.022222, 0.080056]; numerator=13; denominator=450; n_target=9; n_pair=450. - result `H4_H2_HARM_MISSPECIFIED_B9_COMMON_SUPPORT_CI95` / registry `h2.no_repeat.misspecified.b9.harm_C_minus_A`: value=[-0.022222, 0.080056]; numerator=13; denominator=450; n_target=9; n_pair=450. - result `H4_H2_A_MINUS_B_MISDIAGNOSIS_MISSPECIFIED_B9_COMMON_SUPPORT` / registry `h2.no_repeat.misspecified.b9.no_harm_A_minus_B`: value=0.113333; ci95=[0.068889, 0.160000]; numerator=51; denominator=450; n_target=9; n_pair=450. - result `H4_H2_A_MINUS_B_MISDIAGNOSIS_MISSPECIFIED_B9_COMMON_SUPPORT_CI95` / registry `h2.no_repeat.misspecified.b9.no_harm_A_minus_B`: value=[0.068889, 0.160000]; numerator=51; denominator=450; n_target=9; n_pair=450. - result `H4_MCQ_GENERATOR_MINUS_PRODUCTION_MISSPECIFIED_DIAGNOSTIC` / registry `misspecification.item_type.mcq.generator_minus_production`: value=-0.005756; ci95=[-0.007220, -0.004277]; numerator=-869.942501; denominator=152853; n_target=27; n_pair=15400. - result `H4_NUMERIC_GENERATOR_MINUS_PRODUCTION_MISSPECIFIED_DIAGNOSTIC` / registry `misspecification.item_type.numeric.generator_minus_production`: value=0.082579; ci95=[0.080117, 0.085107]; numerator=550.308891; denominator=5238; n_target=13; n_pair=3694.

H5. No outcome estimate was generated because the analysis was excluded pre-outcome; the decision table records this as not evaluated.

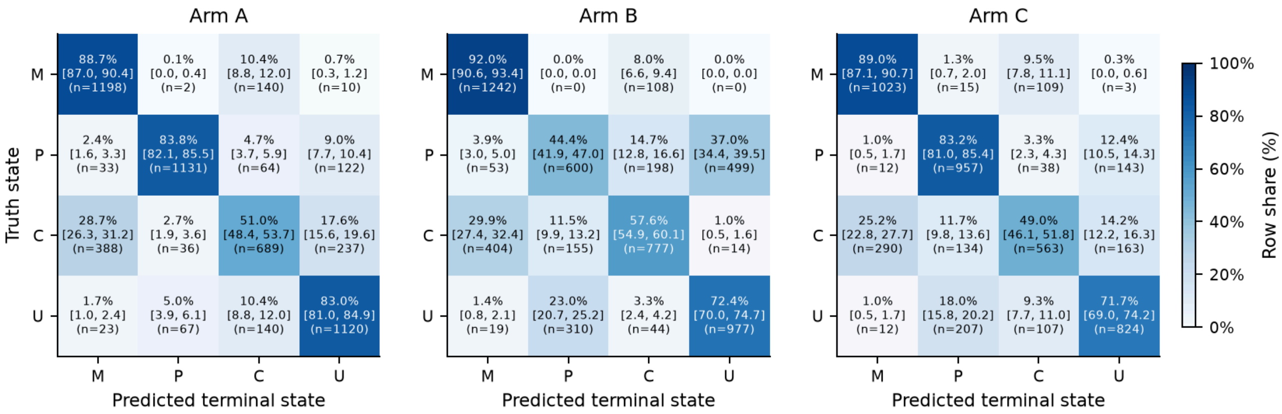
Source artifact: `metric_registry.json` (`sha256:3493b2687cecaa91b0ef7ae25f0dacdaa3ea05a5405c79a4321814d5fa092a7c`). Publication PNG bytes were copied only after contract hash verification.

Verified publication figures



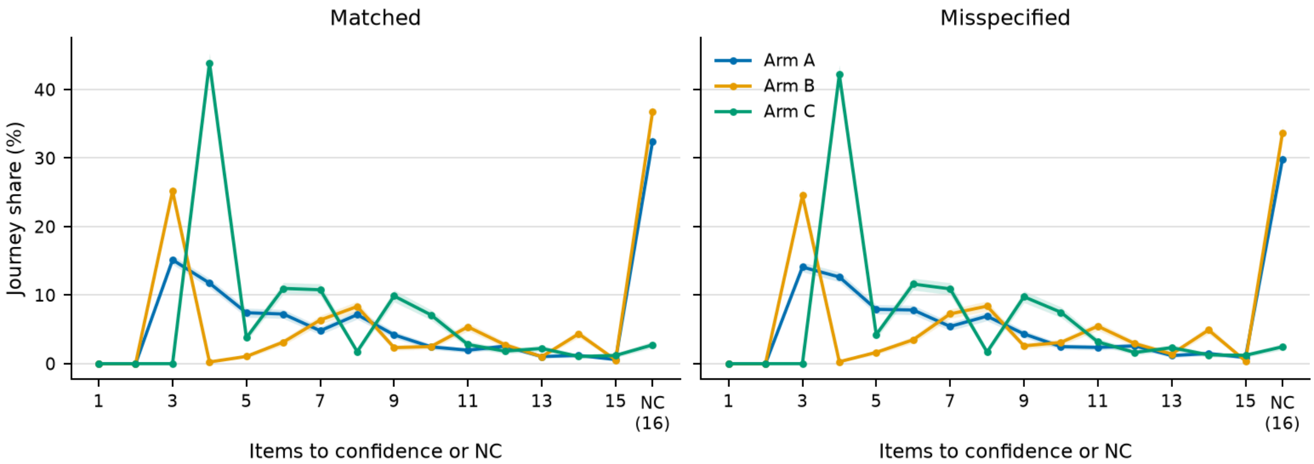
Cells show target-equal row percentage, 95% CI, and raw count; n = 5,400 journeys in Arm A.

Four-state terminal confusion (all valid terminal outputs)



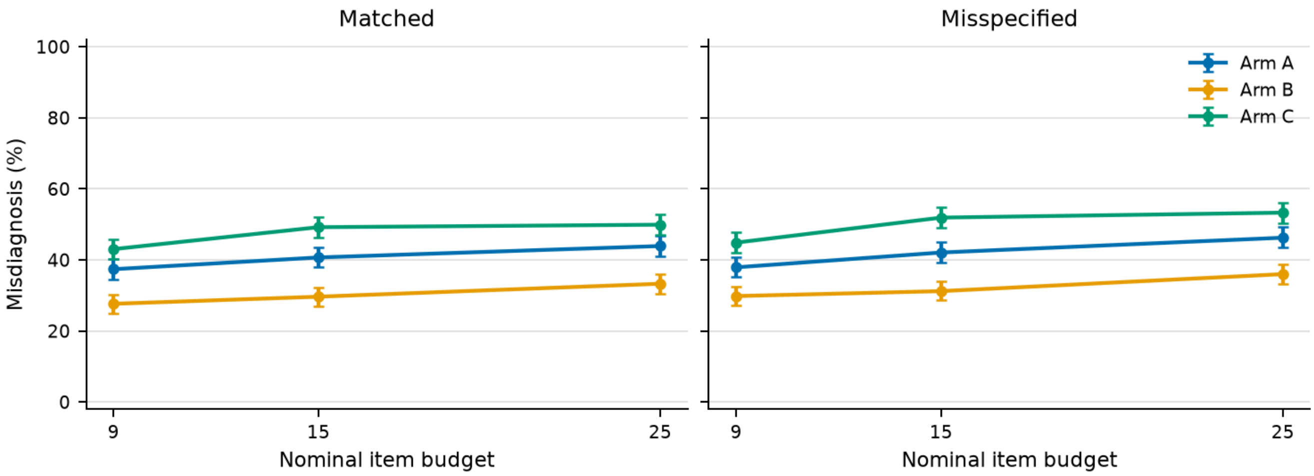
Cells show target-equal row percentage, 95% CI, and raw count; n = 5,400 journeys in Arm A.

Convergence-time distribution at budget 15

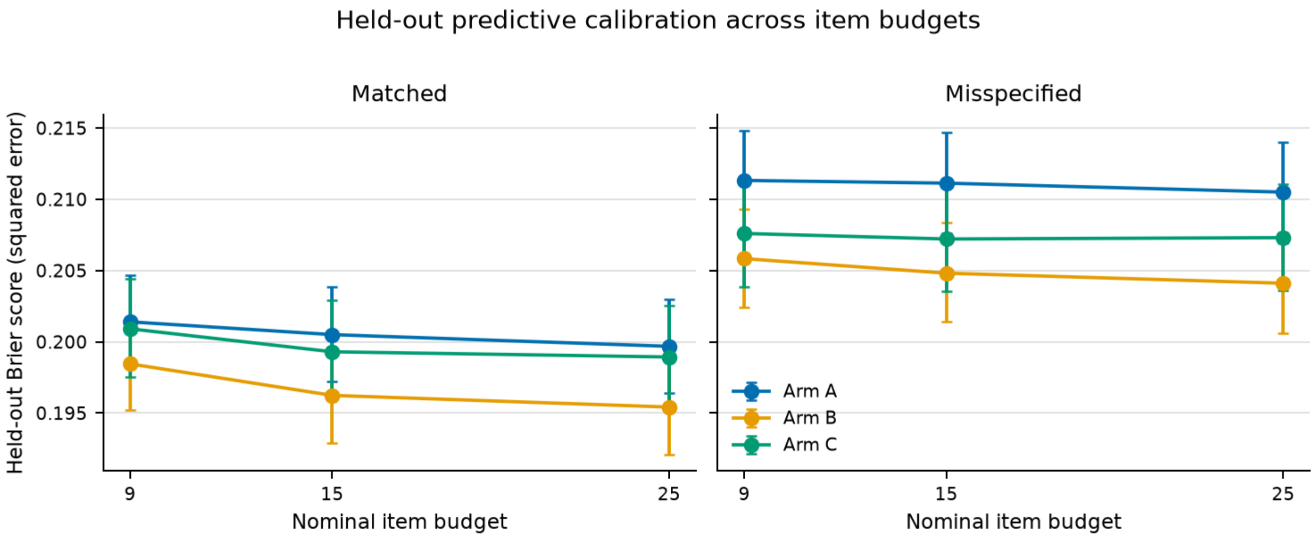


NC = 16 (budget + 1); shaded bands: 95% CI; n = 4,600 journeys per arm-condition.

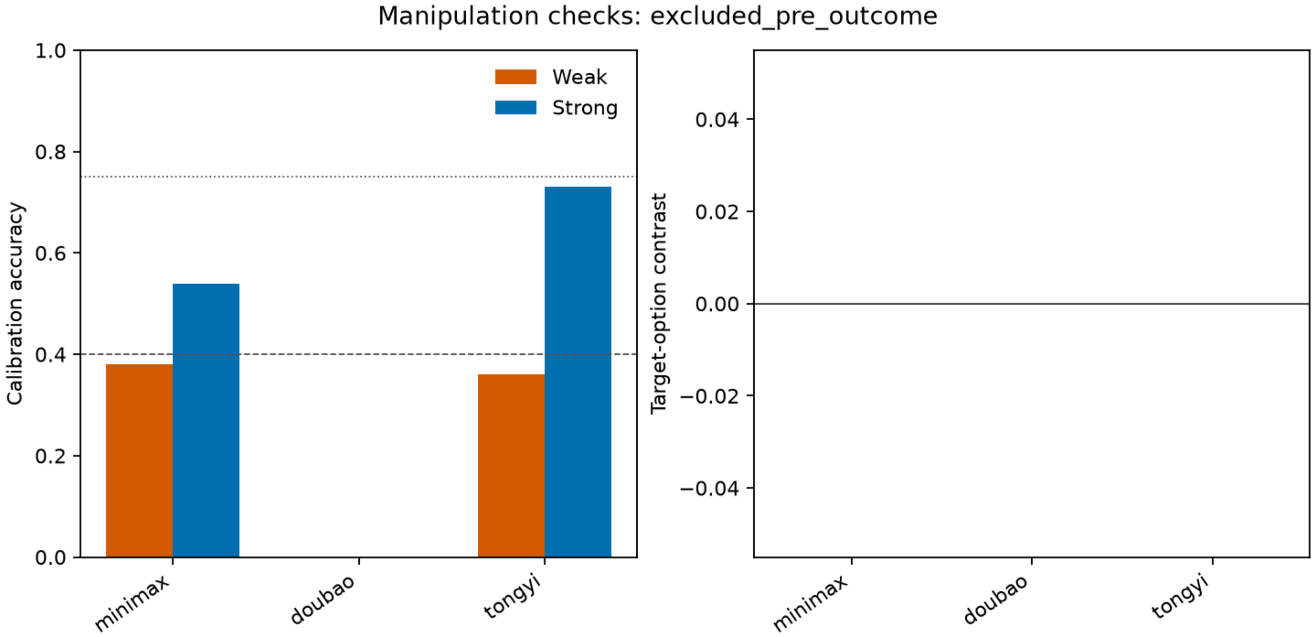
C-state misdiagnosis across item budgets

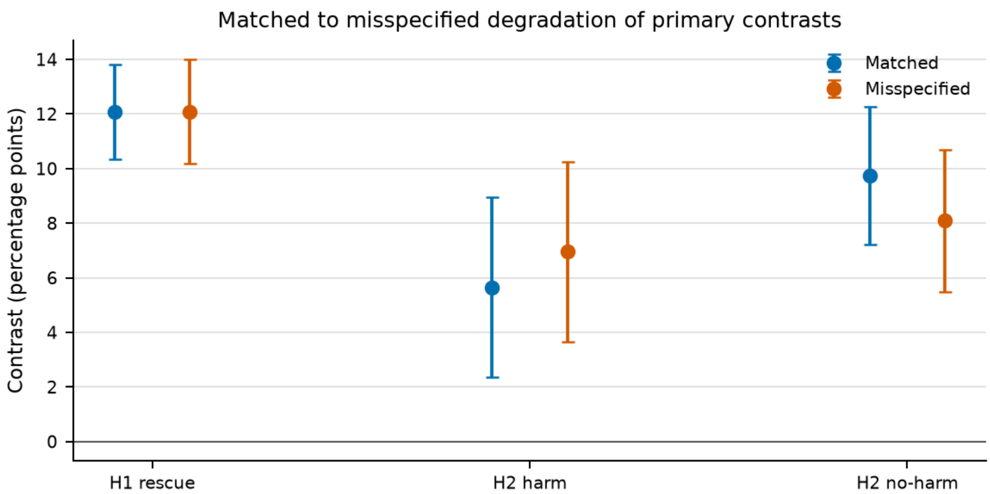


Points are target-equal estimates; error bars: 95% CI; n = 1,150 journeys per arm-budget cell.

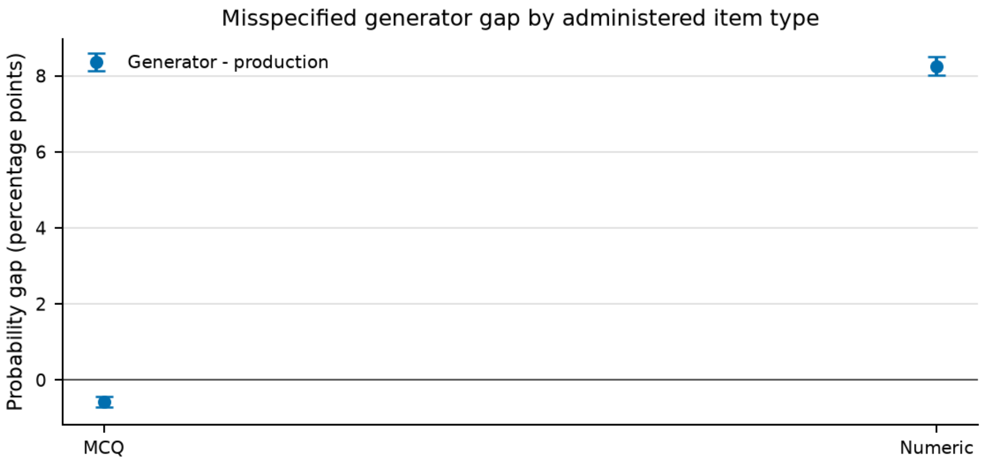


Lower is better; error bars: 95% CI; n = 4,600 journeys/cell. Internal calibration only.

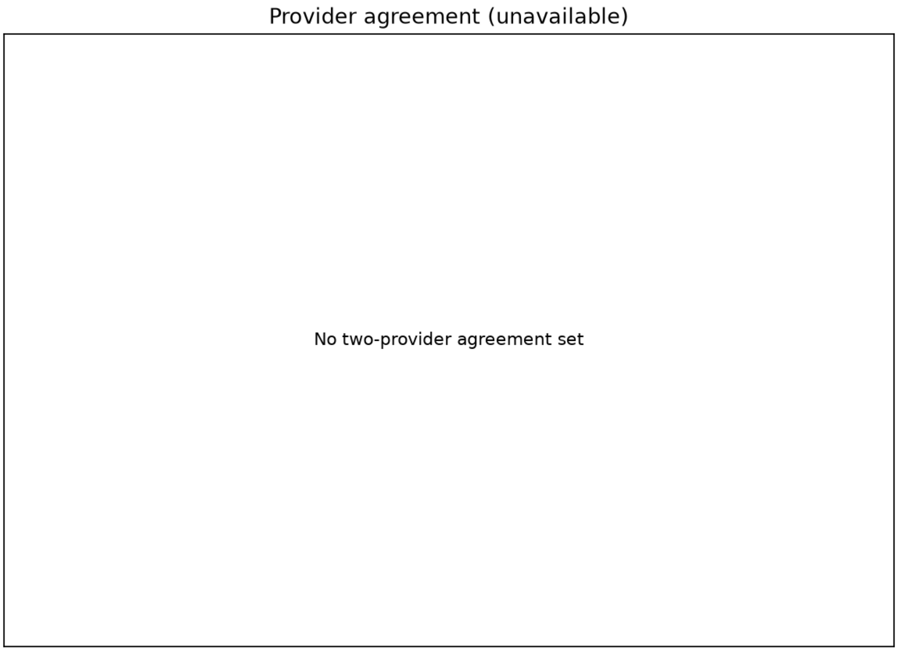


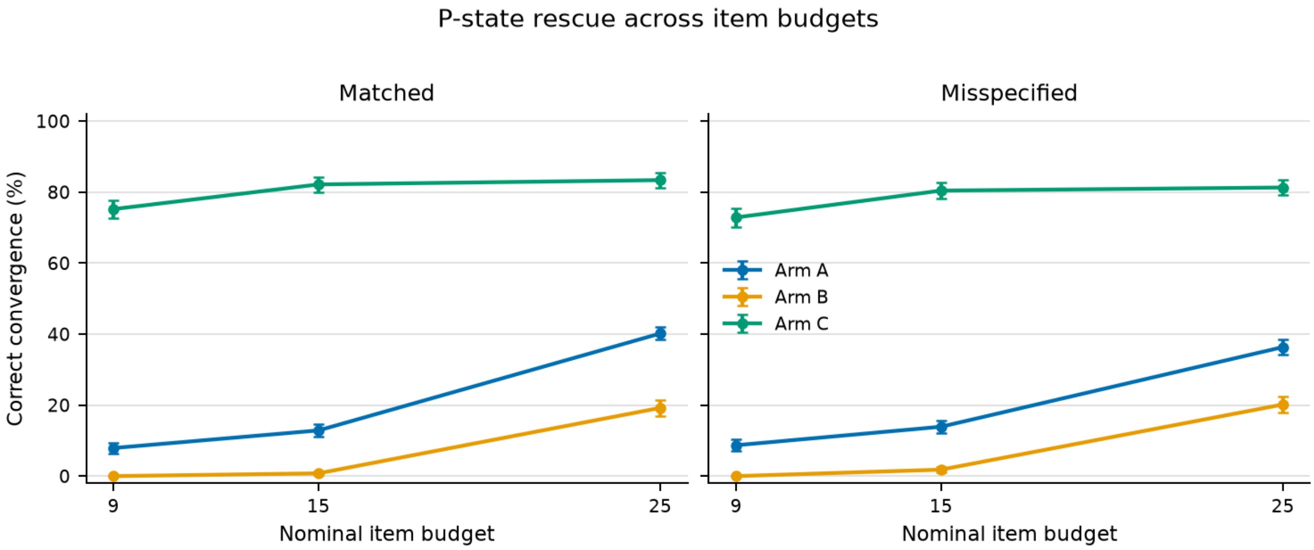


H1 rescue: matched - misspecified = +0.0 pp (95% CI -2.6, +2.5)
H2 harm: matched - misspecified = -1.3 pp (95% CI -6.0, +3.5)
H2 no-harm: matched - misspecified = +1.7 pp (95% CI -2.1, +5.3); n = 1,150 paired journeys/condition.



Misspecified condition; target-stratified 10,000-resample journey-cluster bootstrap; 95% CI.
diagnostic only, not an item-type H1/H2 estimand.
MCQ: 152,853 events / 15,400 journeys / 27 targets.
Numeric: 5,238 events / 3,694 journeys / 13 targets.





Points are target-equal estimates; error bars: 95% CI; n = 1,150 journeys per arm-budget cell.

7. Discussion

H1. The frozen P-state rescue criterion was partially supported. Some required evidence favored rescue, but the full conjunction did not pass; the paper therefore makes no full-support claim.

H2. The criterion was not supported. The fixed-quota harm contrast remained positive, but the belief-triggered arm failed the no-harm requirement relative to the local-only arm; this is not a reversal of the fixed-quota pattern.

H3. The subordinate adaptive sanity check was supported. It remains secondary because adaptive-testing efficiency is established prior art.

H4. Under the frozen misspecification stress, both predicted directions persisted. This result describes sensitivity to one declared perturbation family and cannot establish behavioral realism.

H5. The LLM-persona hypothesis was excluded pre-outcome because its machine mapping gate was unavailable. It was not evaluated and is not reported as a negative finding.

These findings remain limited to simulated, finite-budget diagnosis in the tested catalog. They do not establish structural non-identifiability, human validity, learning gains, or educational efficacy.

8. Contribution Statement and AI Use

This project uses a three-layer responsibility model.

Student responsibility. The student set the product direction, narrowed scope to chemistry, defined the diagnostic-versus-recommendation boundary, requested a re-review with academic value as the priority, and commissioned the frozen v2 route. The decision-layer synthesis proposed the finite-budget state and probe-policy framing; the student approved that route and set the pre-outcome honesty and quality gates. Dated evidence is indexed in `decision_log.md`, especially D11, with repository-local anchors in the `2026-07-13 18:01` overview entry and the frozen v2 brief.

AI-assisted work under review gates. AI systems proposed designs, implemented code and automation, executed the simulation and analysis, drafted the current result interpretation and prose, generated tests, and performed adversarial reviews. Those outputs were constrained by written briefs, frozen contracts, deterministic tests, source-data gates, and post-implementation review. Final interpretation and any submission decision remain pending student review; AI assistance is not represented as independent student authorship.

Audit layer. Git commits, the annotated experiment tag, config and input hashes, seed derivations, isolation attestations, raw simulated envelopes, test suites, and machine-generated result identifiers make the work inspectable. These controls reduce untraceable discretion; they do not turn a simulated study into human evidence.

9. Honest Limitations

1. **Simulation is not human validation.** Programmatic and LLM-simulated students do not establish behavior, learning, usability, or efficacy in real students.
2. **Finite-budget claim only.** P and U differ in production local likelihood and are asymptotically identifiable. The claim is limited to practical non-identifiability at 9, 15, and 25 items under the tested conditions.
3. **Model circularity remains.** The matched generator uses inference probabilities. The misspecified generator tests a declared perturbation family, not all plausible students.
4. **Coverage is narrow.** Only 27 of 135 knowledge-graph nodes are open, and only 23 meet the prerequisite-family gate for H1/H2.
5. **Binary and MCQ-heavy evidence.** The current R5 distractor maps are absent in the confirmatory channel, and much of the catalog has multiple-choice traces. Rich error-process claims are out of scope.
6. **Sparse pools repeat.** Some targets cannot supply 25 independent families. The stress estimand reports repetition, and the no-repeat sensitivity uses a smaller precomputed common-support set.
7. **Static truth is not learning.** The latent truth does not change during a journey. The study evaluates one-session diagnosis, not learning transitions or retention.
8. **Held-out Brier is internal.** Under the matched generator, it checks internal calibration only; it is not an external validity estimate.
9. **LLM manipulation can fail.** Provider and persona exclusions must be reported, and surviving cells do not justify generalizing to human students.
10. **Single project context.** The system was built and audited in one Shanghai chemistry project by one student using extensive AI assistance. Independent replication and domain review remain necessary.

10. Reproducibility

The frozen statistical contract is `experiments/analysis_plan.md`. The canonical configuration is `experiments/config/confirmatory_v1.json`. The runner validates the trusted R5 catalog, 27 open nodes, 23

H1/H2-eligible nodes, four truth states, three arms, two conditions, 50 replicates, and 32,400 planned journeys without writing outcomes.

For the completed programmatic run, the controller committed the reviewed runner, created an annotated experiment tag at that exact HEAD, and bound a truthful UTC run-start value. Execution required `python -B` or `PYTHONDONTWRITEBYTECODE=1`. Raw simulation artifacts remain under `data/sim_store/confirmatory/<run_id>/`; they never enter real student stores. S3 regenerated the programmatic estimates and figures from raw artifacts and populated only `results_contract.md`. The H5 collection is complete but was excluded pre-outcome because the frozen explicit misconception-to-option mapping gate could not be satisfied. Zero providers qualified, and no supported/partially-supported/not-supported H5 outcome decision was made.

Publication, repository push, and DOI actions remain outside this local reproducibility chain and require separate user authorization.

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